Applied Thermal Engineering 174 (2020) 115101

Contents lists available at ScienceDirect

Applied Thermal Engineering

journal homepage: www.elsevier.com/locate/apthermeng

Parametric study of a free-piston Stirling cryocooler capable of providing 350 W cooling power at 80 K

Shunmin Zhu[a,b], Guoyao Yu[a,*], Xiaowei Li[c], Wei Dai[a,b], Ercang Luo[a,b]

[a] Key Laboratory of Cryogenics, Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, Beijing 100190, China
[b] School of Future Technology, University of Chinese Academy of Sciences, Beijing 100049, China
[c] China Ship Development and Design Center, Wuhan 430064, China

HIGHLIGHTS

- Key operating parameters of a high-capacity free-piston Stirling cryocooler were optimized.
- Performance of two different types of ambient heat exchanger were studied.
- An overall efficiency of 26.8% was achieved in experiment after optimization.
- A better performance could be expected with an optimized shell-and-tube type ambient heat exchanger.

ARTICLE INFO

Keywords:
Free-piston
Stirling cryocooler
High cooling capacity
Operating parameter
Heat exchanger

ABSTRACT

High cooling capacity free-piston Stirling cryocooler (FPSC) has been considered a promising candidate for high temperature superconducting and gas liquefaction applications, which require a cost effective and highly reliable cryocooler capable of providing large cooling power (several hundred watts to several kW) at liquid nitrogen temperature. This paper introduced an advance in a FPSC capable of providing 350 W cooling power at liquid nitrogen temperature. To reveal the relationship between operating parameters and cooling performance, the integrated impact of mean pressure and working frequency on acoustic impedance, acoustic field as well as exergy loss of cooler is studied numerically. The numerical results indicate that for the current design, by tuning the mean pressure and working frequency, an optimal operating scenario of the refurbished FPSC could be obtained, which leads to a higher thermal efficiency. Based on the optimization, a series of experiments were carried out on the improved experimental setup. A cooling power of 350 W at 80 K was obtained with an input electric power of 3.57 kW and the corresponding overall relative Carnot efficiency reached up to 26.8%. Compared with previous reports, the performance has been greatly improved by 25.2%. In addition, a special attention is paid to the heat transfer at the ambient heat exchanger due to the large detected local solid-gas temperature difference in the previous work. Comparisons of two different types of ambient heat exchanger in calculations and experiments indicate that an even better performance could further be expected by employing a shell-tube type heat exchanger with 1 mm inner diameter and a porosity around 0.16.

1. Introduction

Ever since the debut of high-temperature superconductors in 1986 [1], significant advances in high temperature superconducting (HTS) technology have been made in the past three decades, with prominent application areas including power transmission cables, electric motors, generators, fault current limiters, and transformers. So far, cooling down HTS applications has mostly been realized by sub-atmospheric boiling of liquid nitrogen, whose temperature limit is the triple point of nitrogen at 63.15 K [2]. Lower operating temperatures allow higher current densities, but require active cooling approaches. Besides temperature, concerns pertaining to the active cooling scheme consist in overall efficiency, reliability, practicability and cost. This motivates the research and development of closed-cycle based cryocooler for the sake of expediting the commercialization of HTS applications.

Currently, qualified cryocoolers can be divided into recuperative-cycle based type (reverse-Brayton, Joule-Thomson and Claude) and regenerative-cycle based type (Stirling, pulse-tube and Gifford-

* Corresponding author.
E-mail address: gyyu@mail.ipc.ac.cn (G. Yu).

https://doi.org/10.1016/j.applthermaleng.2020.115101
Received 23 December 2019; Received in revised form 5 February 2020; Accepted 17 February 2020
Available online 19 February 2020
1359-4311/

Nomenclature

A	cross-sectional area (m^2)
A_{disp}	cross-sectional area of the displacer (m^2)
A_{rod}	cross-sectional area of the rod (m^2)
BL	force factor ($N\ A^{-1}$)
D	piston diameter (mm)
d_h	hydraulic diameter (mm)
e	mass-specific total gas energy ($J\ kg^{-1}$)
f	operating frequency (Hz); darcy friction factor
F	viscous pressure gradient ($N\ m^{-3}$)
h	convective heat transfer coefficient ($W\ m^{-2}K^{-1}$)
i	imaginary unit
I_{max}	maximum current amplitude (A)
k	gas conductivity ($W\ m^{-1}K^{-1}$)
$\boldsymbol{k_e}$	effective gas conductivity ($W\ m^{-1}K^{-1}$)
k_d	displacer equivalent spring stiffness ($N\ m^{-1}$)
K	piston equivalent spring stiffness ($N\ m^{-1}$); total local loss coefficient
K_{gas}	gas spring stiffness ($N\ m^{-1}$)
K_m	mechanical spring stiffness ($N\ m^{-1}$)
L	heat exchanger length (m)
L_e	winding inductance (mH)
m	displacer moving mass (kg)
M	piston moving mass (kg)
Nu	Nusselt number
p_b	pressure wave amplitude in the backside (Pa)
p_{comp}	pressure wave amplitude in the compression space (Pa)
p_{exp}	pressure wave amplitude in the expansion space (Pa)
P	gas pressure (Pa)
Ph(Z)	phase angle of the cooler acoustic impedance (°)
P_r	pressure ratio
P_0	mean pressure (Pa)
q	instantaneous axial heat flux ($W\ m^{-2}$)
Q_c	cooling power (W)
$\boldsymbol{Q_w}$	heat flow per unit length ($W\ m^{-1}$)
R	piston mechanical resistance ($N\ s\ m^{-1}$)
R_e	winding resistance (Ω)
Re(Z)	real part of the cooler acoustic impedance ($Pa\ s\ m^{-3}$)
R_m	displacer mechanical damping coefficient ($N\ s\ m^{-1}$)
$\boldsymbol{S_x}$	wetted perimeter (m)
T	section-average temperature (K)
$\boldsymbol{T_w}$	temperature of negative z surface (K)
u	mean-flow velocity in the x direction ($m\ s^{-1}$)
u_d	displacer velocity ($m\ s^{-1}$)
U_d	volume flow rate at the cold end of the displacer ($m^3\ s^{-1}$)
U_f	volume flow rate at the frontside of the linear compressor ($m^3\ s^{-1}$)
V_b	back space volume of the compressor (m^3)
W_a	acoustic power delivered by the linear compressor (W)
W_e	consumed electric power (W)
W_{exp}	expansion work (W)
W_{loss}	acoustic power dissipated by the mechanical damper (W)
x_d	displacer displacement (m)
X_{max}	maximum piston displacement (mm)
Z	cooler acoustic impedance
γ	specific heat ratio
ε	mass-specific internal gas energy ($J\ kg^{-1}$)
η_{carnot}	Carnot efficiency (%)
η_{cooler}	relative Carnot efficiency of cooler (%)
η_{overal}	overall relative Carnot efficiency (%)
θ_{pU}	phase difference between the pressure wave and the volume flow rate (°)
ρ	gas density ($kg\ m^{-3}$)
ω	angular frequency ($rad\ s^{-1}$)
$\lvert\ \rvert$	magnitude of complex number
$\hat{}$	complex number

McMahon) [3]. Reverse-Brayton and Claude cryocoolers have cooling power range above 10 kW because of the difficulty in downscaling the cold turbo-machinery while maintaining high efficiency. Joule-Thomson cryocooler with mixed refrigerants could provide cooling power down to liquid nitrogen temperature but with low efficiency and bulky size [2]. Commercially available Gifford-McMahon cryocooler can provide hundreds of watts of cooling power below 77 K [4], while the existence of rotary valves and oil lubricated compressor makes it inefficient and unreliable. Pulse tube cryocooler (PTC) might be a good candidate due to its high reliability and potentially high thermal efficiency [5]. However, orientation issue induced by the natural convection within the pulse tube greatly limit its efficiency and thus application [6]. Stirling cryocoolers have shown promising prospects in the field of superconductivity by virtue of high efficiency and compact size.

Stirling cooling engines were firstly made by Alexander Kirk in 1874 [7], while research on Stirling cryocooler was boosted by Philips Company (now renamed as Stirling Cryogenics BV) since 1946, who successfully commercialized it for gas liquefaction in 1956 [8,9]. After the invention of free-piston Stirling engine in the early 1960s, the first development of free-piston Stirling cryocooler (FPSC) was accomplished at the Philips Laboratories [10]. The following significant development includes the invention of split-Stirling cryocooler [11] and Oxford split-Stirling cryocoolers [12], with low cold-tip vibration and long-life span, respectively. Subsequently, some novel configurations including the duplex Stirling refrigerator [13–17], liquid piston Stirling refrigerator [18], diaphragm FPSC [19], and so on have also been proposed.

Generally speaking, Stirling cryocoolers could be categorized as kinematic type, where piston and displacer are mechanically linked to the drive shaft, or free-piston type, where piston and displacer are not constrained by crankshafts or other transmissions. Compared with FPSC, traditional kinematic Stirling cryocooler (KSC) is more mature and have already found application in superconducting facilities. Schlosser et al. [20] reported an HTS transformer run by Siemens. The project aimed at showing the prospects for superconducting railway transformers. A closed cooling cycle with sub-cooled nitrogen was employed to cool down the HTS transformer, and a 4-cylinder KSC (named LPC-4) manufactured by Stirling Cryogenics BV is used to cool the gaseous nitrogen to sub-cooled nitrogen. The LPC-4 has a nominal cooling capacity of 4 kW at 77 K or 2.5 kW at 66 K. Two identical LPC-4 units have also been used in cryogenic refrigeration system of the Albany HTS cable project, which planned to install a 350 m underground HTS cable as part of the power grid of host utility Niagara Mohawk [21,22]. Another HTS cable project conducted by Sumitomo Electric, *etc.* also adopted KSC as the refrigeration source. This project aimed to operate a 66 kV, 200 MVA HTS cable in a real power grid in order to demonstrate its reliability and stable operation. The cooling system provided by Mayekawa Mfg. Co. Ltd. includes six KSCs, each of which with a cooling capacity of 1 kW at 77 K or 0.8 kW at 67 K [23,24].

At the same time, researches and developments focusing on KSC have always been booming. Xu et al. [25] developed a beta-type KSC and studied its operating characteristics, in which the cryocooler is driven by a crank-rod mechanism and can provide a cooling capacity of 700 W at 77 K with an electric input power of 10.9 kW, corresponding to an overall relative Carnot efficiency (defined as $\eta_{overal} = Q_c/W_e/\eta_{carnot}$, where Q_c, W_e, and η_{carnot} denote the cooling power, consumed electric power, and Carnot efficiency, respectively) of 18.2%. Cheng et al. [26] developed a prototype of beta-type Stirling cryocooler with rhombic driver mechanism and an efficient thermodynamic model were built simultaneously for predicting the performance of the cryocooler. The experimental results indicated that the no-loading temperature of the cryocooler can reach 93 K at 1000 rev min^{-1} with 0.3 MPa helium. A non-ideal adiabatic analysis aiming at modeling a gamma-type Stirling refrigerator thermodynamically was done by Batooei and Keshavarz [27], and a multi-objective optimization by design of experiment method was also performed. Hachem et al. [28] established a thermodynamic model considering the thermal losses that occur in the regenerator and performed a parametric study of an air filled beta-type Stirling refrigerator under different operating and geometrical parameters. Li and Grosu [29] used an isothermal model which considers various losses to analyze the performance of a KSC used in the Mars Express project by the European Space Agency. Energy and exergy analysis indicate that mechanical friction is the largest mechanical loss and conduction loss is the largest heat loss. Then, the effect of various parameters on the cryocooler performance was also studied based on the developed model. Finally, the model was validated on the PPC-102 KSC [30].

Although the KSC has been extensively studied and applied for the past several decades, issues of gas contamination and leakage arising from the rigid transmission make frequent maintenance indispensably required. The advent of FPSC delicately circumvents these problems by acoustically tuning the movement of displacer and piston. FPSC therefore possesses a number of important advantages including high reliability, zero maintenance, high thermal efficiency, and compact size [31].Over the past three decades, low cooling capacity (≤ 10 W) FPSCs in the liquid nitrogen temperature range have been well studied and rapidly developed, mainly for military and space applications [32,33].

Up-scaling of a FPSC to hundreds of watt of cooling power at liquid nitrogen temperature involves new challenges. For example, as similarly occurring in a PTC, the length-to-diameter ratio of the regenerator in a FPSC decreases with increasing cooling capacity. The large regenerator diameter increases the inclination of gas flow maldistributions, inhomogeneous transverse temperature distribution, and the relating acoustic field distortion associated with the tactics. The interaction of these effects deteriorates the cooling performance severely. Another issue is the equivalent loss pertinent to the large-diameter displacer, which on one hand consumes the recovered acoustic power and on the other hand cause the untune of the acoustic field.

In order to meet the HTS requirements, a number of studies aiming at developing FPSC with high cooling capacity at liquid nitrogen temperature range have been carried out in recent years. In 2011, Karandikar and Fiedler from Superconductor Technologies Inc. conducted a feasibility study, focusing on a conceptual FPSC capable of providing 650 W cooling power at 65 K, to determine whether the design approach of their existing FPSC named *Sapphire* (with a cooling capacity of 5 W at 77 K) could be scaled to provide larger cooling power and meanwhile preserve the key advantages [34]. Though the results were encouraging, there are no further reports on their test results to date. In 2014, Penswick et al. reported a low-cost, high-capacity FPSC with a cooling capacity of 650 W at 77 K with an input electric power of 5800 W, corresponding to a relative Carnot efficiency of about 30% [35]. In order to provide an efficient cryocooler for the cooling system of an HTS motor in a joint project, Sumitomo Heavy Industries, Ltd., firstly developed and reported a high-efficiency Stirling type PTC in 2013 [36]. While its efficiency was inadequate to meet the demand of the HTS motor. Then they redesigned the expander of the cryocooler and developed two split Stirling cryocoolers, within which the second one with a better cooling performance capable of providing 151 W cooling power at 70 K, i.e., 23.4% of relative Carnot efficiency with an input electric power of 2.15 kW [37,38]. In 2015, Ko *et al.* built a FPSC with a dual-opposed moving magnet linear compressor aimed at cooling HTS devices and pre-cooling a hydrogen liquefier. Their test results indicated that with an input electric power of 8.1 kW, the cryocooler capable of providing 650 W cooling power at 76.8 K. Correspondingly, the relative Carnot efficiency is 21.5% [39,40]. Most recently, a 4 kW-class Stirling cryocooler system including eight such Stirling cryocooler units were tested in conjunction with a superconducting cable demonstration facility [41]. Caughley et al. from Callaghan Innovation of New Zealand explored the concept of diaphragm FPSC which employs metal diaphragms to suspend the displacer. Two prototypes were constructed and the second one with better cooling performance achieved 29 W cooling power at 77 K [19]. Subsequent CFD analysis shows that the configuration has much room for improvement [42].

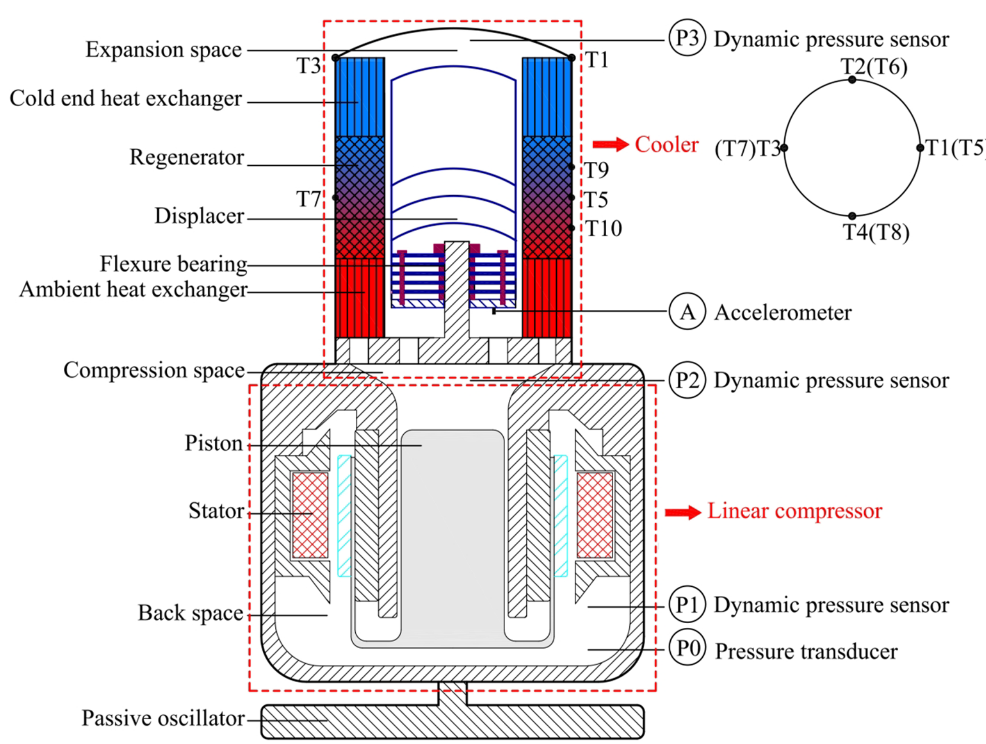


Fig. 1. Cutaway schematic of the FPSC prototype.

Our research group launched the study on high-capacity FPSC in 2014. Theoretical design and calculation on a FPSC with a cooling capacity of 300 W at 80 K from the perspective of thermoacoustic theory were firstly carried out by Yu *et al.* [43]. Initial prototype test indicated the cooler could only provide a cooling power of 56 W at 80 K [44]. Subsequently, key parameters that affect the cooler acoustic field including displacer moving mass, regenerator porosity and operating frequency were fully investigated and analyzed focusing on local impedance, phase difference and their relationship with cooler cooling performance [45]. After a deep insight into the acoustic field characteristics and structural optimization, some improvements were made on the initial structure. Both the flexure bearings fixing approach and the displacer cylinder fixing approach were redesigned to ensure a uniform clearance seal and the equivalent mechanical damping coefficient of displacer was reduced by a factor of 10 accordingly. These revisions made the cooling performance of the refurbished FPSC improve substantially.

Although several high-capacity FPSC prototypes working at liquid nitrogen temperature have been successfully constructed in recent years, very few detailed studies have been carried out on the effects of operating parameters and ambient heat exchanger (AHX) despite that both of which are indispensable for further improvement of a FPSC cooling performance. Therefore, there is an urge to investigate the impact of different operating parameters such as mean pressure and operating frequency on a specific FPSC and unveil the common principles behind it. Besides, commonly used configurations of an AHX in a high-capacity FPSC should also be compared comprehensively to choose suitable configuration and geometric parameters. Therefore, in this work, based on the refurbished FPSC prototype, an integral impact of operating parameters including mean pressure and operating frequency on acoustic impedance, acoustic field as well as exergy loss of cooler was investigated to choose an optimal set of parameters. Subsequently, experiments were conducted to validate the calculations, showing encouraging improvement over the previous work. Additionally, a special attention is paid to heat transfer at the AHX due to a large detected local solid–gas temperature difference. The influence of two different types of AHX on the performance of a high-capacity FPSC was fully investigated. Finally, some conclusions are drawn.

2. Numerical simulations

2.1. System configuration

Fig. 1 presents the cutaway schematic of our FPSC prototype. The FPSC has a gamma-type configuration (i.e., piston and displacer are confined in different cylinder), and it consists of a moving-magnet type linear compressor and a cooler. The compressor adopts a single piston configuration, and the piston is supported by a gas bearing mechanism. The casing vibration is balanced by a passive oscillator installed at the bottom of the compressor housing. The detailed parameters of the linear compressor are listed in Table 1.

As for the cooler, it includes a compression space, an AHX, a regenerator, a cold end heat exchanger (CHX), an expansion space, and a displacer. Table 2 shows the structural parameters of each component. Regenerator is one of the key components of a FPSC, in which acoustic power is consumed to transfer entropy from the CHX to the AHX, thereby obtaining a cooling effect. In a small cooling capacity design, the regenerator is usually deployed inside the displacer for compactness. While for a high-capacity design, an annular arrangement of the regenerator that encompasses the displacer is favored for less constraints on size. The regenerator housing is filled with 900 pieces of stainless-steel screens of 300 meshes, with a porosity of 0.76. Displacer is another key component in a FPSC, which is used to recover the expansion work and adjust the acoustic field inside the regenerator to ensure high thermal efficiency. For the prototype, flexure bearings are employed to support the displacer and provide a gas clearance seal of several micrometers. As shown in Fig. 1, the outer rim of the flexure bearings is bolted with the displacer. In addition, both the AHX and CHX are of radial-fin type and made of copper, and the AHX is cooled by 25 °C recirculating chilling water.

Table 2
Structural parameters of the cooler.

Component	Specifications	
Expansion space	Volume (cc)	45
CHX	Type	Radial-fin
	Equivalent diameter (mm)	90
	Length (mm)	30
	Fin gap (mm)	0.3
	Fin radial height (mm)	19
Regenerator	Type	Wire-mesh
	Equivalent diameter (mm)	90
	Length (mm)	60
	Screen mesh count	300
	Screen mesh wire diameter (μm)	31
	Porosity	0.76
Displacer	Displacer diameter (mm)	75
	Rod diameter (mm)	25
	Moving mass (kg)	1.03
	Spring constant (kN/m)	88.2
AHX	Type	Radial-fin
	Equivalent diameter (mm)	90
	Length (mm)	35
	Fin gap (mm)	0.5
	Fin radial height (mm)	16.25
	Porosity	0.26
Compression space	Volume (cc)	300

2.2. Numerical model

In order to further improve the cooler thermal efficiency based on previous work [45], an optimization has been conducted by studying the effect of key operating parameters including mean pressure and operating frequency. A numerical model of the cooler was established accordingly by using Sage software, which was developed by Gedeon Associates [46]. With a graphical interface, Sage models each cooler component individually based on its dimensions, geometry and material. The cooler components are connected by mass flow, pressure wave, force and energy flow, thus the optimization can be accomplished. The one-dimensional governing equations for momentum, continuity, and energy in the gas domain are listed as follows,

$$\frac{\partial \rho A}{\partial t} + \frac{\partial \rho u A}{\partial x} = 0 \tag{1}$$

$$\frac{\partial \rho u A}{\partial t} + \frac{\partial u \rho u A}{\partial x} + \frac{\partial P}{\partial x}A - FA = 0 \tag{2}$$

$$\frac{\partial \rho e A}{\partial t} + P\frac{\partial A}{\partial t} + \frac{\partial}{\partial x}(u\rho e A + uPA + q) - Q_w = 0 \tag{3}$$

Table 1
Parameters of the linear compressor.

BL (N/A)	R_e (Ω)	L_e (mH)	M (kg)	K_m (kN/m)	R (N·s/m)	D (mm)	I_{max} (A)	X_{max} (mm)
55	0.15	192.75	7	181.2	70	118	50	14.2

where P, u, A and ρ denote the working gas pressure, mean-flow velocity in the x direction (longitudinal), cross sectional area and working gas density, respectively; $e = \varepsilon + u^2/2$is mass-specific total gas energy, ε is mass-specific internal gas energy; F is viscous pressure gradient (for a heat exchanger with hydraulic diameter d_h and length L, it can be formulated in terms of Darcy friction factor f and total local loss coefficient K, $F = -(f/d_h + K/L)\rho u\ |u|/2)$; q is instantaneous axial heat flux, which can be formulated as $q = -k_e \frac{\partial T}{\partial x} A$, k_e is the effective gas conductivity; $Q_w$is the heat flow per unit length through the negative z surface, which can be calculated as

$$Q_w = hS_x(T_w - T) = Nu(k/d_h)S_x(T_w - T) \tag{4}$$

where Nu is the Nusselt number, k is gas conductivity, S_x is the wetted perimeter, $T_w$is the temperature of negative z surface, and T is the section-average temperature.

Besides the momentum, continuity, and energy equations, the momentum equation of the displacer is required, which can be written as

$$p_{exp}A_{disp} - p_{comp}(A_{disp} - A_{rod}) - R_m u_d - k_d x_d = m\frac{d^2x_d}{dt} \tag{5}$$

where p_{exp} and p_{comp} denote the pressure wave in the expansion space and compression space, respectively; A_{disp} and A_{rod} denote the cross-sectional areas of the displacer and rod, respectively; R_m is the displacer mechanical damping coefficient; u_d, x_d and m are the velocity, displacement and moving mass, respectively, of the displacer; and k_d is the equivalent spring stiffness, which consists of two parts, i.e., the mechanical spring stiffness and gas spring stiffness.

Together with boundary conditions established by pressure source, moving components and heat exchanger temperature or heat inflow/outflow, Eqs. (1)–(3) and Eq. (5) form the basis for simulating the cooler.

2.3. Operating parameters optimization

Generally speaking, to rule off the influence of re-assembling, a most effective way to adjust the acoustic impedance of a cooler is to change the operating parameters. In this section, the characteristics of the cooler operating at different frequencies and pressures has been investigated in detail to find the optimum operating conditions for maximum cooling efficiency. Therefore, a wide pressure range from 2.5 MPa to 3.5 MPa and a wide frequency span from 45 Hz to 55 Hz have been studied simultaneously. Besides, the acoustic power delivered by the compressor is kept constant at 2959 W (a cooling capacity of 350 W at 80 K was achieved in experiment with this measured amount of acoustic power) and the cooling temperature is fixed at 80 K.

2.3.1. Internal acoustic field analysis

According to our previous work [45], the internal acoustic field influences the cooling performance of a FPSC to a great extent. Therefore, some crucial acoustic parameters which are closely related to the intrinsic losses of the cooler are calculated and analyzed firstly.

Fig. 2(a) and (b) show the calculated real part and phase angle of the cooler acoustic impedance (denoted by Re(Z) and Ph(Z), respectively. Z is the ratio of complex pressure wave amplitude to complex volume flow rate amplitude) as a function of operating frequency and mean pressure, respectively. For a fixed operating frequency, both Re (Z) and Ph(Z) increase gradually with the rise of mean pressure. In contrary, the decrease of operating frequency also increases Re(Z) and Ph(Z). While for the same operating frequency span (i.e., the operating frequency decreases from 55 Hz to 45 Hz), the higher the mean pressure, the larger the increment of Re(Z) but the smaller the increment of Ph(Z). According to the calculated results, the effect of mean pressure on Re(Z) and Ph(Z) is weaker than the effect of operating frequency on the Re(Z) and Ph(Z).

Fig. 3(a) and (b) show the regenerator inlet volume flow rate amplitude and pressure wave amplitude as a function of operating frequency and mean pressure, respectively. When the operating frequency of FPSC is kept constant, Re(Z) descends with the decreasing mean pressure, thus when transferring a fixed acoustic power, the volume flow rate amplitude is obviously larger since the pressure wave amplitude slightly decreases with the decreasing mean pressure, as shown in Fig. 3(a) and (b). In addition, operating frequency also plays an important role in affecting the regenerator inlet volume flow rate amplitude and pressure wave amplitude. All in all, the lower the mean pressure and the operating frequency, the higher the regenerator inlet volume flow rate amplitude. The higher the mean pressure and operating frequency, the lower is the regenerator inlet volume flow rate amplitude.

The contour graph in Fig. 3(c) show the regenerator inlet mass flow rate amplitude as a function of operating frequency and mean pressure. When the operating frequency is lower than 53 Hz, the regenerator inlet mass flow rate increases linearly with the increasing mean pressure. When the operating frequency is higher than 53 Hz, the regenerator inlet mass flow rate first decreases slightly and then almost keeps unchanged with the increasing mean pressure. Meanwhile, operating frequency also has a significant effect on the regenerator inlet mass flow rate. A minimum regenerator inlet mass flow rate is achieved at an operating frequency of 52.5 Hz and a mean pressure of 2.5 MPa.

2.3.2. Cooling performance analysis

The above section illustrates the dependence of the internal acoustic field on the variation of the mean pressure, and the following analysis will reveal the intrinsic relationship between internal acoustic field and

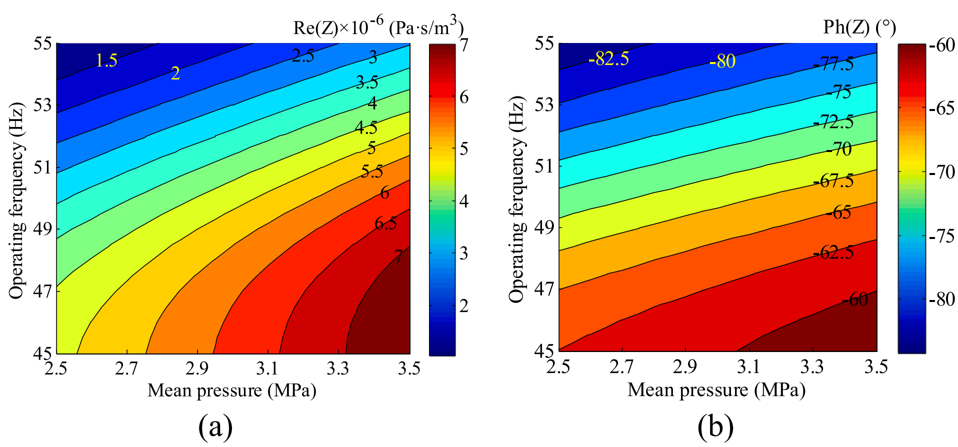


Fig. 2. Calculated (a) Re(Z) and (b) Ph(Z) as a function of operating frequency and mean pressure.

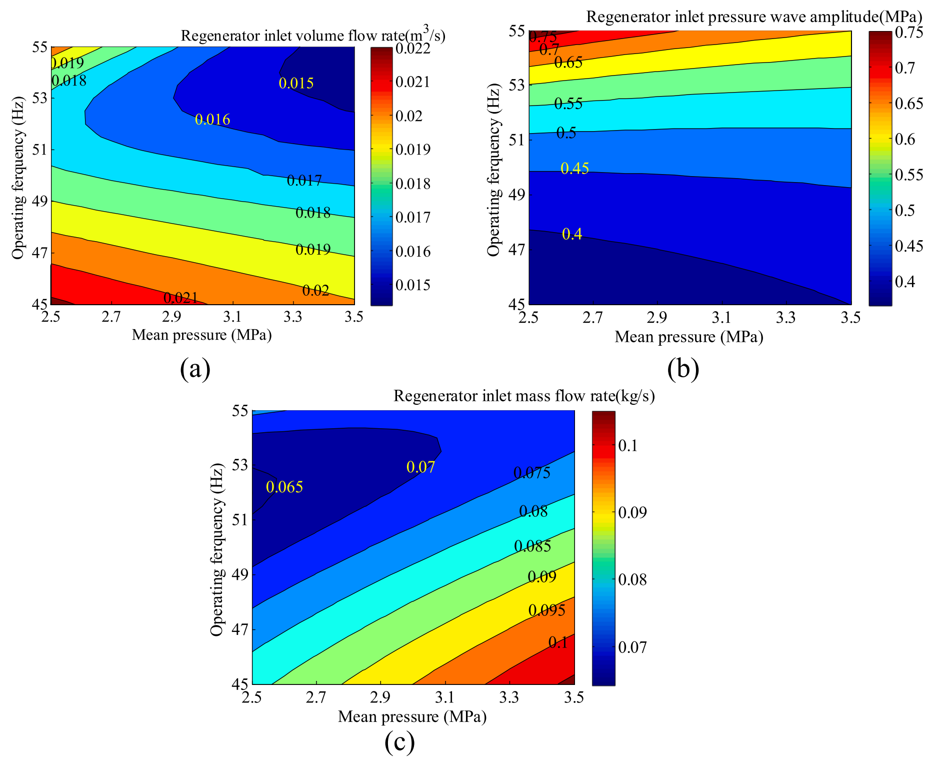


Fig. 3. Regenerator (a) inlet volume flow rate, (b) inlet pressure wave amplitude and (c) inlet mass flow rate amplitude as functions of operating frequency and mean pressure.

performance of the cooler through the method of exergy analysis.

Fig. 4(a) shows the influence of mean pressure on the cooling power at 80 K at different operating frequency. When the operating frequency is lower than 51 Hz, the cooling power decreases slightly with the increasing mean pressure. Besides, the cooling power first increases and then decreases slowly with the increasing mean pressure when the operating frequency is between 51 Hz and 53 Hz. When the operating frequency is higher than 53 Hz, the cooling power decreases

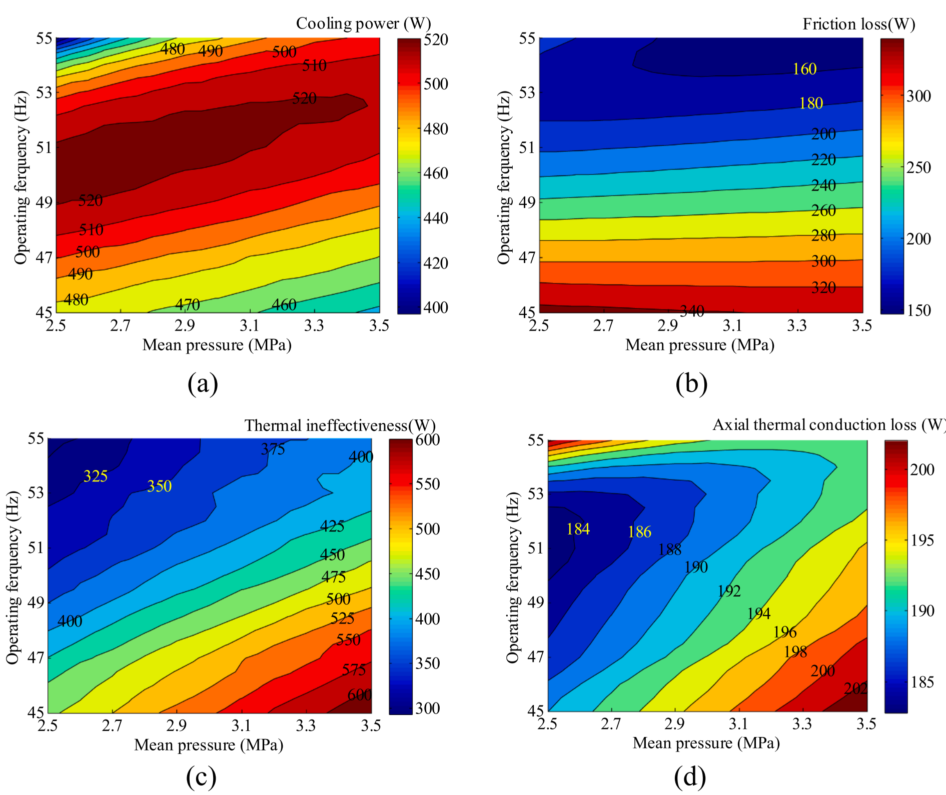


Fig. 4. (a) Effects of mean pressure on (a) cooling power at 80 K, (b) regenerator friction loss, (c) regenerator thermal ineffectiveness and (d) regenerator axial thermal conduction loss at different operating frequency.

dramatically with the increasing mean pressure. A maximum cooling power of 526 W is achieved at an operating frequency of 51 Hz and a mean pressure of 2.75 MPa.

Exergy analysis is an effective method to evaluate the irreversibility of each component in a FPSC. According to our previous study [45], most exergy loss in a FPSC occurs in the regenerator, so we firstly calculated the loss of the regenerator. Roughly speaking, the loss of the regenerator consists of the friction loss, thermal ineffectiveness, and axial thermal conduction, which are induced by the viscous flow friction, the internal irreversible heat transfer, and the axial thermal conduction (including working gas, regenerator solid matrix and regenerator wall), respectively.

Fig. 4(b)–(d) illustrate the effect of mean pressure on regenerator friction loss, thermal ineffectiveness and axial thermal conduction at different operating frequency, respectively. The crucial influence factor of the friction loss is the operating frequency, which determines the viscous penetration depth of the working gas. The lower the operating frequency, the larger the viscous penetration depth, which in turn leads to the rise of the friction loss. In contrast, mean pressure has lesser effect on the regenerator friction loss. When the operating frequency is lower than 53 Hz, the regenerator friction loss almost keeps constant with the increasing mean pressure. While the regenerator friction loss decreases slightly with the increasing mean pressure once the operating frequency is higher than 53 Hz. Quite different from friction loss, both mean pressure and operating frequency play an important role in regenerator thermal ineffectiveness. For a constant operating frequency, the higher the mean pressure, the larger the thermal ineffectiveness. This is because that the increase of mean pressure causes the increase of regenerator inlet mass flow rate (as shown in Fig. 3(c)). For thermal conduction loss that includes both the gas and solid sides, the increase of mean pressure leads to a tiny increase of effective thermal conductivity (integration of longitudinal thermal conduction loss divided by the multiplication of the overall temperature gradient and flow area) of helium, thus the thermal conduction induced by gas increases accordingly. The axial thermal conduction loss caused by regenerator wall and solid matrix is related to the temperature gradient across the regenerator, and both of them remain relatively constant with the change of mean pressure and working frequency. To sum it up, mean pressure has little effect on thermal conduction loss. For example, when the operating frequency is lower than 52 Hz, as the mean pressure increases from 2.5 MPa to 3.5 MPa, the corresponding increment ratio of axial thermal conduction losses is around 7.0%.

Fig. 5(a) gives the effect of mean pressure on regenerator total loss at different operating frequency. Since the thermal ineffectiveness loss of the regenerator accounts for the largest proportion of the regenerator total loss, the variation trend of the total loss is similar to that of thermal ineffectiveness loss, i.e., for a constant operating frequency, the regenerator total loss increases with increasing mean pressure. Obviously, this loss gradient decreases dramatically when the operating frequency increases from 45 Hz to 55 Hz. The change in other components' total loss (including losses occurred in the compression space, AHX, CHX, expansion space, and displacer) with mean pressure and operating frequency is opposite to that of regenerator total loss. As shown in Fig. 5(b), the higher the operating frequency and the lower the mean pressure, the larger the other components total loss. A combination of Fig. 5(a) and (b) indicates that there exists an optimal operating frequency and mean pressure to minimize the total loss and maximize the cooling power of the cooler. According to Fig. 4(a), the optimal operating frequency is around 51 Hz and the optimal mean pressure is around 2.75 MPa.

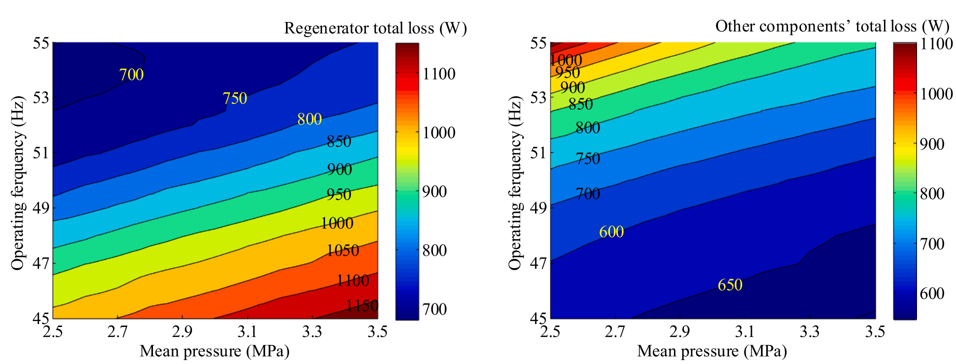
2.4. Effect of AHX

AHX in a FPSC is somewhat inconspicuous since that its function is to transport time-averaged transversal heat from the working gas to the circulating cooling water. As a matter of fact, besides the basic heat rejection, AHX also exerts impact on the acoustic impedance. Generally speaking, there are mainly two typical configurations of the AHX employed in a FPSC, i.e., radial-fin type (as shown in Fig. 6(a)) and shell-and-tube type (as shown in Fig. 6(b)). In this section, the effect of the heat exchanger key parameters including porosity (the ratio of the cross-sectional area of the helium flow channel to the total cross-sectional area of the heat exchanger) and hydraulic diameter on the cooling performance in the two different geometric configurations AHX are numerically studied. In simulations, the acoustic power delivered by the compressor is kept constant at 2959 W. Meanwhile, the operating frequency and the mean pressure are set to be 52 Hz and 2.75 MPa, respectively.

2.4.1. Radial-fin type AHX

Firstly, the radial-fin type AHX is investigated. The fin gap and fin radial height were kept constant at 0.5 mm and 16.25 mm in the calculations, and the AHX porosity is changed by varying the fin thickness.

The magnitude of exergy loss of the AHX clearly reflects the heat transfer capability of the AHX. Fig. 7(a) shows the exergy loss of AHX versus the AHX porosity under different hydraulic diameter. Obviously, the majority of the total exergy loss of the AHX is the thermal ineffectiveness between the working gas and the copper fins. When the hydraulic diameter is 1 mm and the AHX porosity is 0.05, the thermal ineffectiveness is nearly 290 W, which occupies 65.6% of a total AHX exergy loss of 442 W. When the AHX porosity further increases to 0.5, the thermal ineffectiveness is 297 W, which amounts to 62.3% of a total AHX exergy loss of 477 W. In addition, the increase of the hydraulic diameter will significantly increase the thermal ineffectiveness and decrease the friction loss slightly especially when the AHX porosity is small. For example, when the AHX porosity is 0.05 and the hydraulic diameter increases from 1 mm to 2 mm, the thermal ineffectiveness increases from 290 W to 535 W, and the friction loss declines from 62 W



(a) (b)

Fig. 5. Effects of mean pressure on (a) regenerator total loss and (b) other components' total loss at different operating frequency.

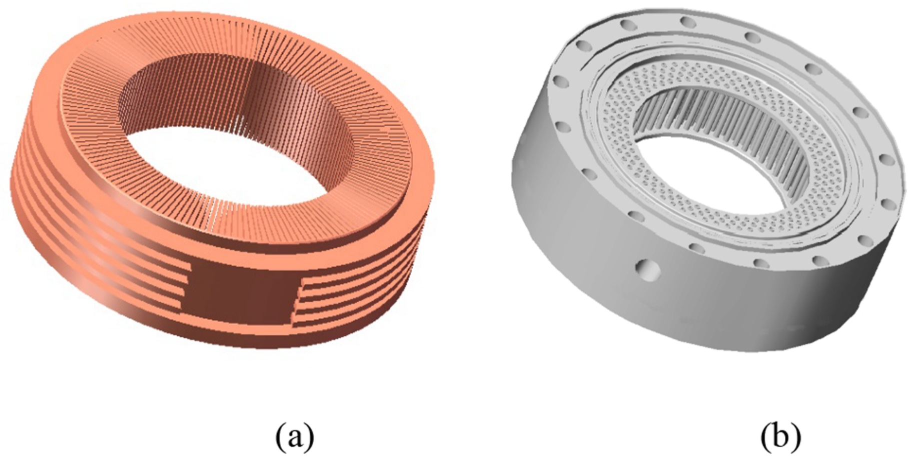


Fig. 6. Configurations of AHXs. (a). radial-fin type; (b) shell-and-tube type.

to 41 W accordingly. What's more, hydraulic diameter has smaller effect on radial thermal conduction loss, compared with that of the fin height and porosity.

Fig. 7(b) shows the cooling power at 80 K and relative Carnot efficiency of the cooler (defined as $\eta_{cooler} = Q_c/W_a/\eta_{carnot}$, where W_a denotes the consumed acoustic power) versus the AHX porosity under different hydraulic diameter when radial-fin type AHX is employed. When the hydraulic diameter is set to be 1 mm, as the AHX porosity rises from 0.05 to 0.5, both the cooling power and relative Carnot efficiency peak, and the cooler can provide a maximum cooling power of 504 W at 80 K, corresponding to a relative Carnot efficiency of 46.4% when the AHX porosity is around 0.21. When a larger hydraulic diameter of 2 mm is selected, the cooling performance deteriorates considerably due to the degradation of the gas–solid heat transfer.

2.4.2. Shell-and-tube type AHX

Similarly, the performance of the shell-and-tube type AHX is analyzed from the perspective of exergy loss. In the calculations, porosity of the shell-and-tube type AHX is changed by varying the tube number. As shown in Fig. 8(a), when compared with the radial-fin type AHX, the total exergy loss is greatly reduced in the shell-and-tube type AHX. In addition, the radial thermal conduction loss is smaller compared with that in the radial-fin type AHX.

Fig. 8(b) shows the cooling power at 80 K and relative Carnot efficiency of the cooler versus the shell-and-tube type AHX porosity under different hydraulic diameter. As shown, for a fixed AHX porosity, a decrease of the hydraulic diameter leads to an increase of the cooling power at 80 K and relative Carnot efficiency due to the decrease of thermal ineffectiveness as shown in Fig. 8(a). More clearly a maximum cooling power at 80 K and relative Carnot efficiency is reached within the range of porosity from 0.05 to 0.5 for either hydraulic diameter. When an optimal shell-and-tube AHX with a porosity of 0.16 and a hydraulic diameter of 1 mm is employed, the cooler can achieve a maximum efficiency of 49.3% and a maximum cooling power of 535 W, which are better than that with a radial-fin type AHX.

3. Experimental investigations and results

3.1. Experiment setup

Based on the simulation and optimization, experiments were carried out aiming at validating the effectiveness of operating parameters optimization and studying the effect of AHX. A photograph of the experimental setup is shown in Fig. 9. The system includes a linear compressor and a cooler. The structural parameters of the cooler are the same as listed in Table 2.

In the experiments, the cold parts are placed inside a vacuum chamber to reduce the thermal radiation loss, and the vacuum environment is maintained by a turbo-molecular pump. In addition, to reduce the radiation heat leakage, external walls of the regenerator and the CHX are all wrapped with multilayer insulation films. For the measurement system, the circumferential wall temperatures of the CHX are monitored by four calibrated PT-100 platinum resistance thermometers (T1–T4, as shown in Fig. 1) with an accuracy of ±0.1 K, and another four PT-100 platinum resistance thermometers (T5–T8, as shown in Fig. 1) of the same type are evenly attached at the

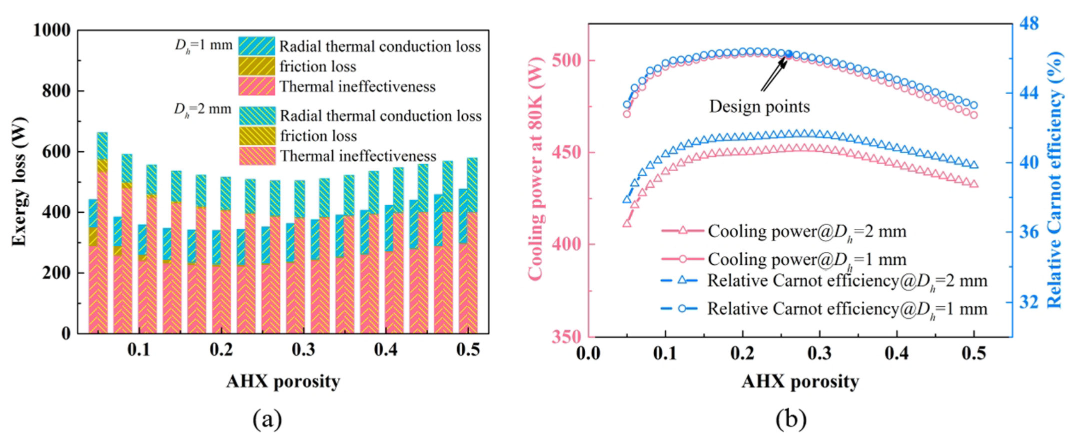


Fig. 7. (a) AHX exergy loss, (b) cooling power at 80 K and relative Carnot efficiency of the cooler versus AHX porosity under different hydraulic diameter when radial-fin type AHX is employed.

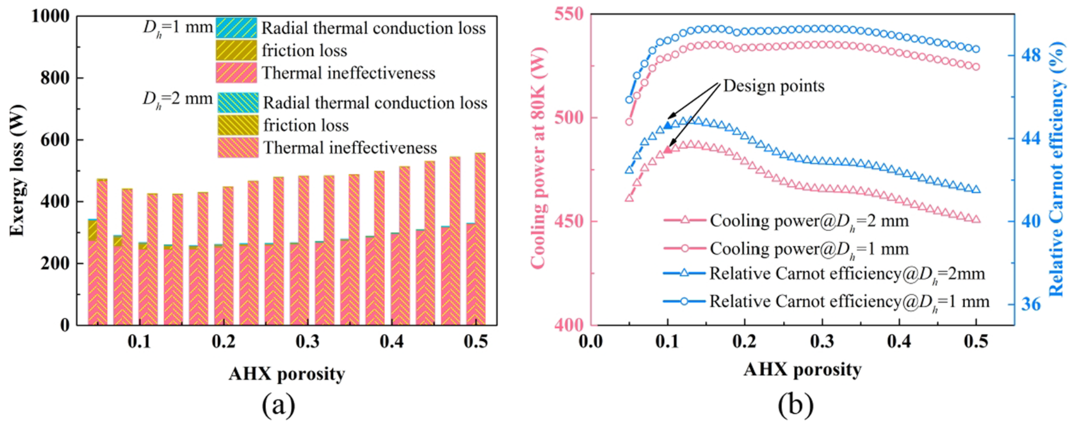


Fig. 8. (a) AHX exergy loss, (b) cooling power at 80 K and relative Carnot efficiency of the cooler versus AHX porosity under different hydraulic diameter when shell-and-tube type AHX is employed.

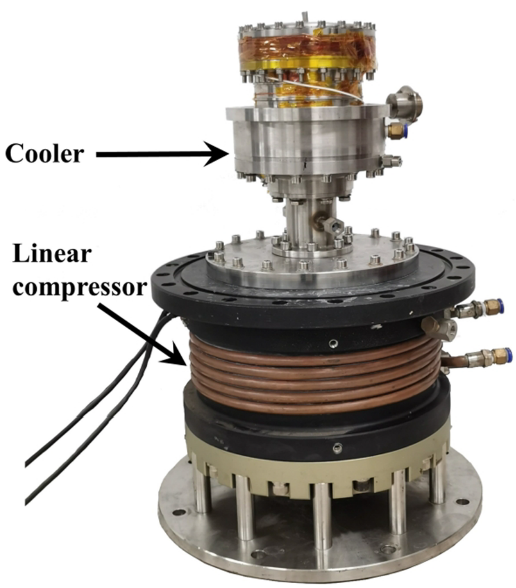


Fig. 9. The photograph of the experimental setup.

circumference to measure the circumferential wall temperatures of the middle of the regenerator. In addition, the lateral wall temperatures at the 1/4 and 3/4 points of the regenerator are measured using another same two PT-100 platinum resistance thermometers (T9 and T 10, as shown in Fig. 1), which have a same radial orientation as T1 and T5.

The gas temperature in the AHX is monitored using a type K thermocouple. Meanwhile, two dynamic pressure sensors (with model 113B21) from PCB Piezotronics Inc. are used to measure local dynamic pressure wave at the backside of the linear compressor and compression space, respectively. Thus the acoustic power delivered by the linear compressor is calculated as

$$W_a = \frac{1}{2}|\hat{p}_{comp}||\hat{U}_f|\cos\theta_{pU} = \frac{\omega V_b \cos\theta_{pU}}{2\gamma P_0}|\hat{p}_b||\hat{p}_{comp}| \tag{6}$$

where U_f is the volume flow rate at the frontside of the linear compressor,θ_{pU} is local phase difference between the pressure wave and the volume flow rate, γ is the specific heat ratio, P_0 is the system mean pressure, V_b is the back side volume of the compressor, and p_b is the pressure wave amplitude in the backside. A cryogenically compatible dynamic pressure sensor (with model 112A05) and an accelerometer (with model M353B14) both from PCB Piezotronics Inc. are employed to measure the dynamic pressure wave at the expansion space and displacer stroke, respectively. Thus the volume flow rate at the cold end of the displacer can be written as $\hat{U}_d = i\omega A_{disp}\hat{x}_d$, and the expansion work can be calculated as

$$W_{exp} = \frac{1}{2}|\hat{p}_{exp}||\hat{U}_d|\cos\theta_{pU} \tag{7}$$

A pressure transducer from Collihigh is used to monitor the mean pressure of the system. Cooling power is measured through three constantan wires which are wound circumferentially on the outside surface of the CHX and heated by a DC voltage source. In addition, the input voltage and current of the linear compressor are measured by a voltage

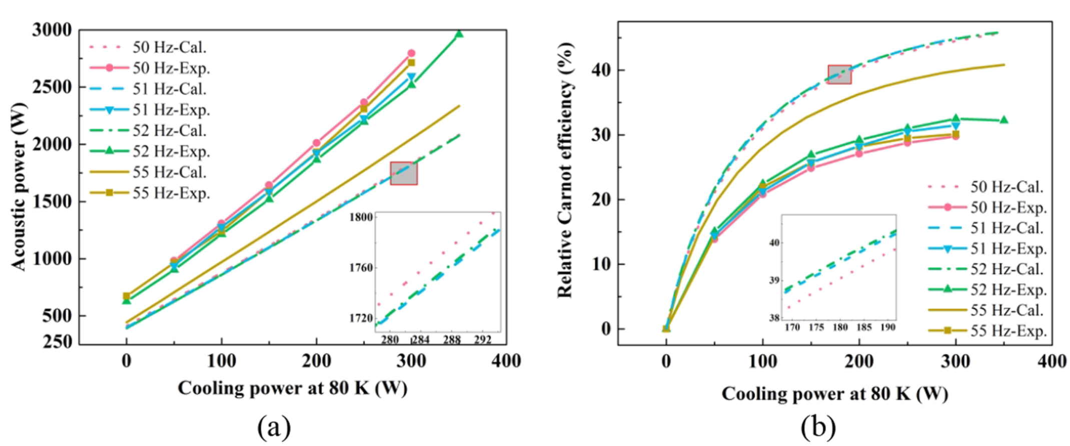


Fig. 10. Dependence of the (a) compressor output acoustic power and (b) cooler relative Carnot efficiency on the cooling power at 80 K at different operating frequency.

differential probe from Tektronix (model P5200A) and a high-precision current probe from PINTECH (model PT-2710), respectively, to calculate the consumed electric power. All signals are simultaneously collected by a dynamic signal acquisition card (model PXI4472B) provided by National Instruments.

3.2. Optimize verification

Firstly, the effect of operating frequency on the system performance is explored. In the experiments, four operating frequencies, i.e., 50 Hz, 51 Hz, 52 Hz and 55 Hz are investigated. Fig. 10 shows the dependence of the compressor output acoustic power and cooler relative Carnot efficiency on the cooling power at 80 K at different operating frequency. With the increase of the cooling power, the consumed acoustic power increases linearly, and the cooler relative Carnot efficiency increases with a decreasing slope both in simulation and experiment. Besides, the cooler consumes the fewest acoustic power to provide an identical cooling power at 80 K when the operating frequency is 52 Hz, thus a maximum cooler relative Carnot efficiency is achieved at 52 Hz, which is in good accordance with the optimization results illustrated in Section 2.3.

Obviously, the calculations show close agreement with the experiments in both tendencies and magnitudes, and the deviation between the experimental and the numerical results could mainly be attributed to the following reasons: (1) The quasi-one-dimensional Sage model employs empirical formulas in calculating flow losses and heat transfer, which causes under-estimation of various losses. Taking a working frequency of 51 Hz as an example, as shown in Fig. 11(a)–(c), due to the under-estimation of losses occurred in the regenerator, the calculated regenerator inlet volume flow rate is much lower than that in experiments especially when the cooling power is much higher as shown in Fig. 11(a), and the pressure drop across the regenerator in experiments is higher than that in simulations (as shown in Fig. 11(b)). This causes the under-estimation of the regenerator consumed acoustic power in simulations (as shown in Fig. 11(c)). (2). Due to manufacture and assembly quality, the displacer parameters like mechanical damping coefficient and spring stiffness deviates from design values to some degree, causing the movement of the displacer unexpected. For example, as shown in Fig. 11(d), the measured displacer displacements are lower than predicted values when the cooling power is much higher. (3). An inhomogeneous lateral temperature distribution around the circumference of the regenerator was observed in experiments, which was probably induced by non-uniform gas flow therein and hence degraded the cooler cooling performance to some extent. It should be noted that this is related to nonlinear effect and could not be captured in the current quasi-one-dimension model. (4) In experiments, gas temperature inside the AHX goes up dramatically with increasing cooling power (as shown in Fig. 18), indicating that the heat exchange capability of the radial-fin type AHX is insufficient.

Then, the effect of mean pressure on the system performance is explored. In the experiments, two different mean pressures, i.e., 2.75 MPa and 3.0 MPa are investigated. Fig. 12 shows the dependence of the compressor output acoustic power and cooler relative Carnot efficiency on the cooling power at 80 K at an optimum operating frequency of 52 Hz. In the simulation, for a constant cooling power, mean pressure affects the consumed acoustic power and relative Carnot efficiency of the cooler slightly. While in experiments, the effect of mean pressure enlarges. The higher the mean pressure, the more the consumed acoustic power for obtaining an identical cooling power, corresponding to a lower cooler relative Carnot efficiency. For a fixed

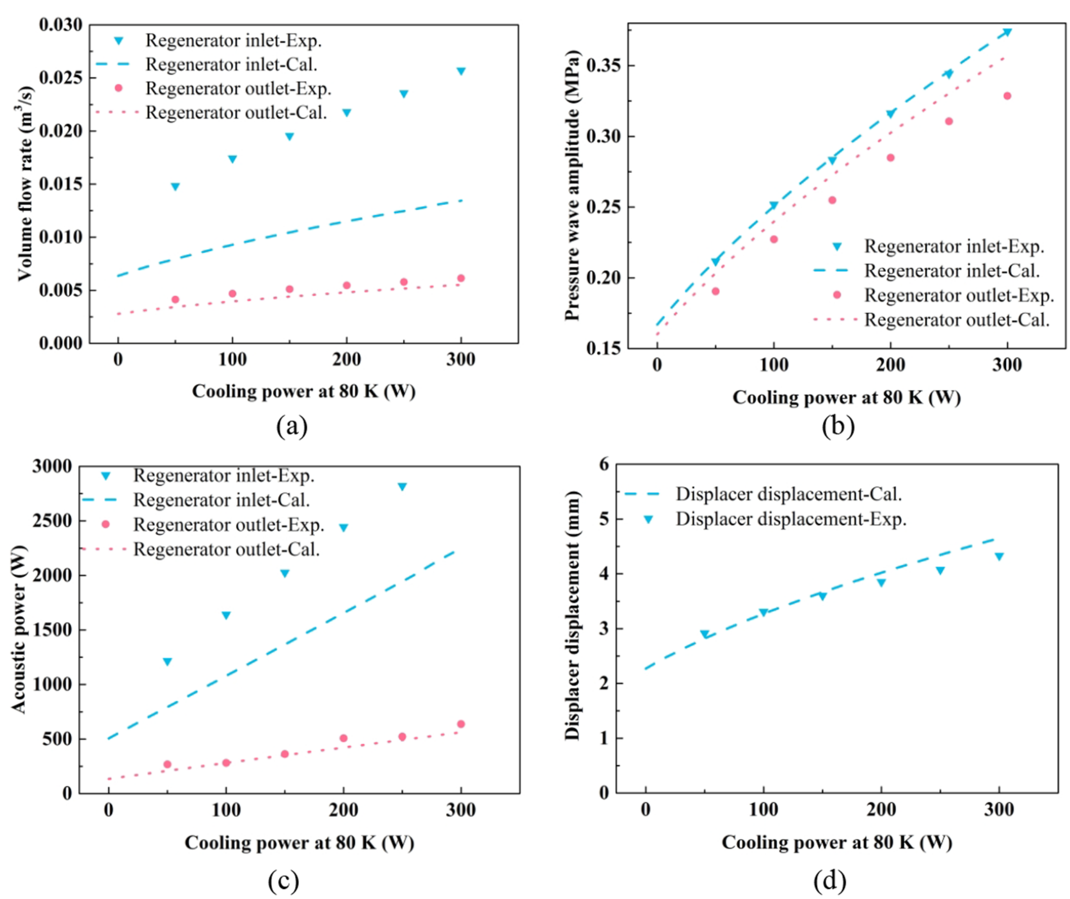


Fig. 11. Dependence of the (a) volume flow rate, (b) pressure wave amplitude and (c) acoustic power at the inlet and outlet of regenerator, and (d) displacer displacement on the cooling power.

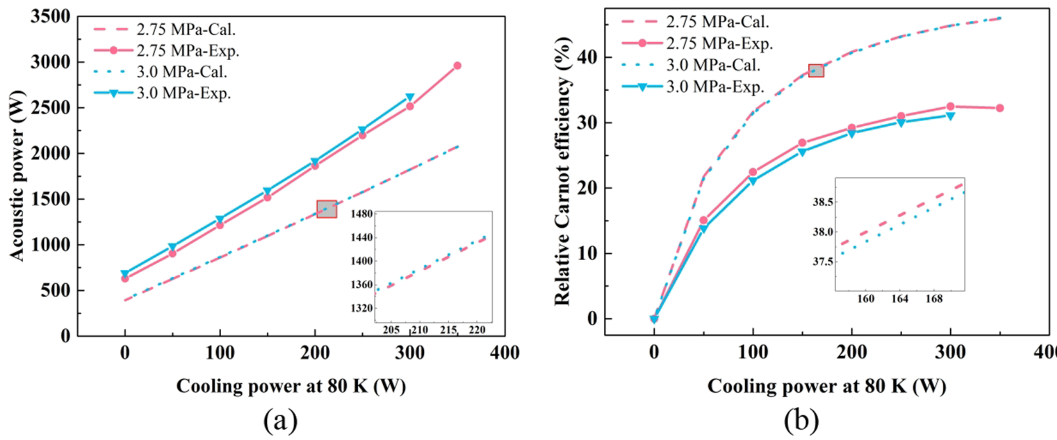


Fig. 12. Dependence of the (a) compressor output acoustic power and (b) cooler relative Carnot efficiency on the cooling power at 80 K.

cooling power at 80 K, a better cooling performance was achieved with a mean pressure of 2.75 MPa, which verifies the effectiveness of the above optimization. The reasons causing the discrepancy between the experimental and the numerical results are the same as that behind the discrepancy in Fig. 9.

Fig. 13 further gives the dependence of the consumed electric power and relative Carnot efficiency of the FPSC (cooling power/electric power consumed by the linear compressor/Carnot efficiency) on the cooling power at 80 K under 52 Hz. The variation trend of the consumed electric power and overall relative Carnot efficiency with increasing cooling power is similar to that of cooler consumed acoustic power and relative Carnot efficiency, respectively. When the mean pressure is 2.75 MPa, the FPSC capable of providing 350 W cooling power at 80 K, i.e., 26.8% of overall relative Carnot efficiency with an input electric power of 3.57 kW.

Fig. 14 shows the maximum temperature difference among the four thermometers on the CHX under different cooling power. It's much clear that the maximum temperature difference is less than 0.5 K, showing good temperature uniformity on the CHX. This might partly be ascribed to the high thermal conductivity of copper and partly imply that flow in the CHX is quite uniform.

Fig. 15 gives the lateral wall temperature distribution in the middle of the regenerator with a mean pressure of 2.75 MPa and a working frequency of 52 Hz. As shown, mean temperature of the measured four thermometers increases slightly with the increase of input electric power. When the input electric power rises from 0.77 kW to 3.57 kW, mean temperature increases from 188.23 K to 195.87 K, and the maximum temperature difference among the four thermometers increases from 12.52 K to 17.87 K. Compared with the reported temperature fluctuation in a high-cooling capacity PTC [6,47], the results in this work are rather encouraging.

3.3. Comparison of two different types of AHX

In order to verify the prediction on the AHX, a series of experiments are conducted by employing an off-the-shelf shell-and-tube type AHX to investigate the influence of AHX on the cooling performance of the FPSC. The dimensions of the shell-and-tube type AHX comprising 200 parallel stainless steel tubes, each of which has an inner diameter of 2.0 mm, outer diameter of 2.8 mm, and a length of 36 mm. The corresponding porosity of the AHX is around 0.10.

Fig. 16 presents the dependence of the consumed electric power and overall relative Carnot efficiency on cooling power at 80 K for the two type of AHXs. The mean pressure is 2.75 MPa, and two different working frequencies, i.e., 50 Hz and 52 Hz are investigated. As shown, a better cooling performance is obtained by adopting the radial-fin type AHX, especially at large input electric power. When the cooling power is small, the system performance of these two AHXs is pretty close; however, with larger input electric power, the radial-fin type AHX obviously outperforms the shell-and-tube type AHX. Taking a cooling power of 300 W under 52 Hz as an example, the FPSC with the radial-fin type AHX achieves a relative Carnot efficiency of 26.39% with an input electric power of 3.10 kW, whereas system with the shell-and-tube type AHX consumed an electric power of 3.62 kW and only achieves an efficiency of 25.4%. As a matter of fact, this slightly lower performance of the shell-and-tube configuration is not surprising as indicated on the design points of Figs. 7(b) and 8(b): for a fixed input acoustic power of 2959 W, the radial-fin type AHX with a porosity of 0.26 and a hydraulic diameter of 1 mm theoretically could provide a cooling power of 502 W at 80 K, while the shell-and-tube type AHX with a porosity of 0.10 and a hydraulic diameter of 2 mm theoretically could only provide a cooling power of 484 W at 80 K. Likewise, it's interesting to see the impact of frequency once the cooling power is kept constant. As shown in Fig. 15, working frequency affects the consumed acoustic power and relative Carnot efficiency of the FPSC slightly when the shell-and-tube type AHX is employed. In contrast, in the system with the radial-fin type AHX, the effect of working frequency enlarges.

Although the current radial-fin type AHX outperformed the shell-and-tube type AHX, both suffered a large intrinsic thermal ineffectiveness, which was verified in experiments by monitoring the gas temperature at the inlet of the AHXs. Meanwhile, pressure ratio (defined as$P_r = (P_0 + p_{comp})/(P_0 - p_{comp})$) at the inlet of the AHX was measured to evaluate the friction loss. Fig. 17 shows the dependence of pressure ratio on consumed acoustic power, and Fig. 18 gives the measured gas

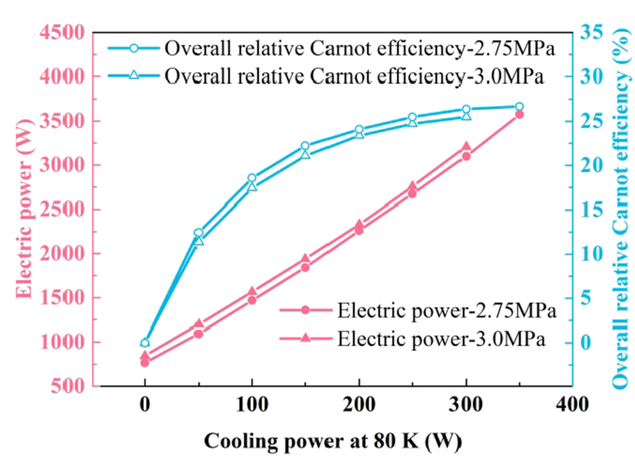


Fig. 13. Compressor consumed electric power and overall relative Carnot efficiency versus cooling power at 80 K.

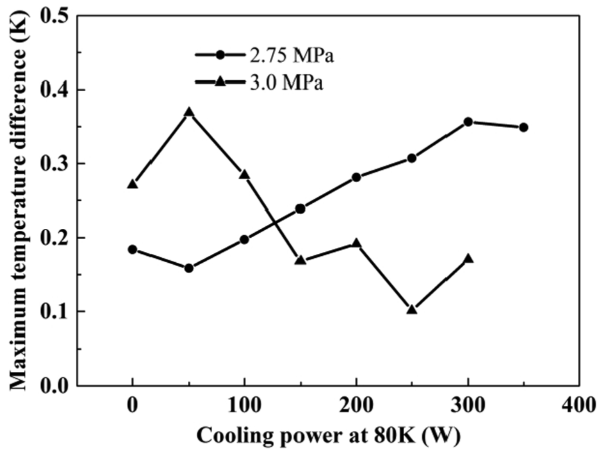


Fig. 14. The maximum temperature difference among the four thermometers on the CHX under different cooling power.

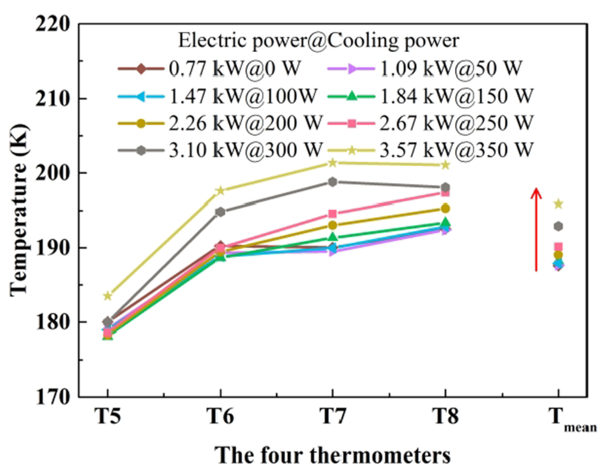


Fig. 15. Lateral wall temperature distribution at the middle of the regenerator.

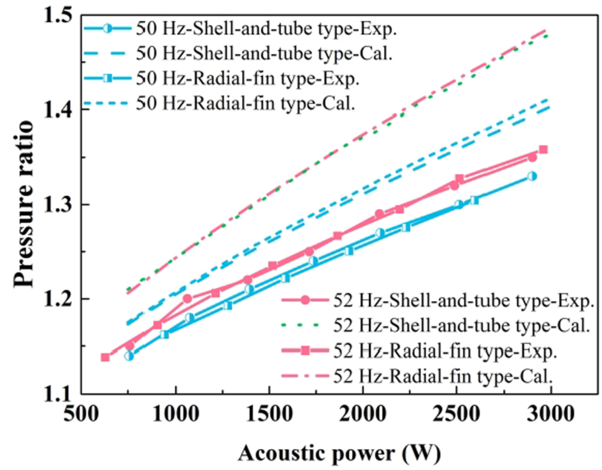


Fig. 17. Dependence of pressure ratio on consumed acoustic power.

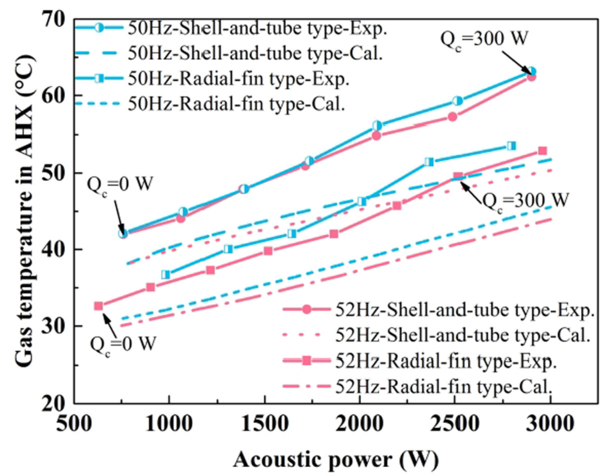


Fig. 18. Dependence of gas temperatures at the inlet of the AHXs on consumed acoustic power.

temperatures at the inlet of the AHXs. For a fixed operating frequency, both the pressure ratio and gas temperatures at the inlet of the AHXs increase almost linearly with the increasing acoustic power. Taking an operating frequency of 52 Hz as an example, when the cooling power is zero at 80 K, the gas temperatures for the two AHX designs were 32 °C for the radial-fin and 42 °C for the shell-and-tube design, indicating temperature differences between the gas and the cooling water of merely 7 °C and 17 °C, respectively (the cooling water temperature is fixed at 25 °C in experiments). When the cooling power increases to 300 W, the gas temperature climbs up to 49 °C for the radial-fin type AHX and 62 °C for the shell-and-tube type AHX, corresponding to a gas–water temperature difference of 24 °C and 37 °C, respectively. According to the numerical calculation in Section 2.3, reducing the tube inner diameter of the shell-and-tube type AHX to 1 mm and increasing the tube number simultaneously to an optimum porosity may improve

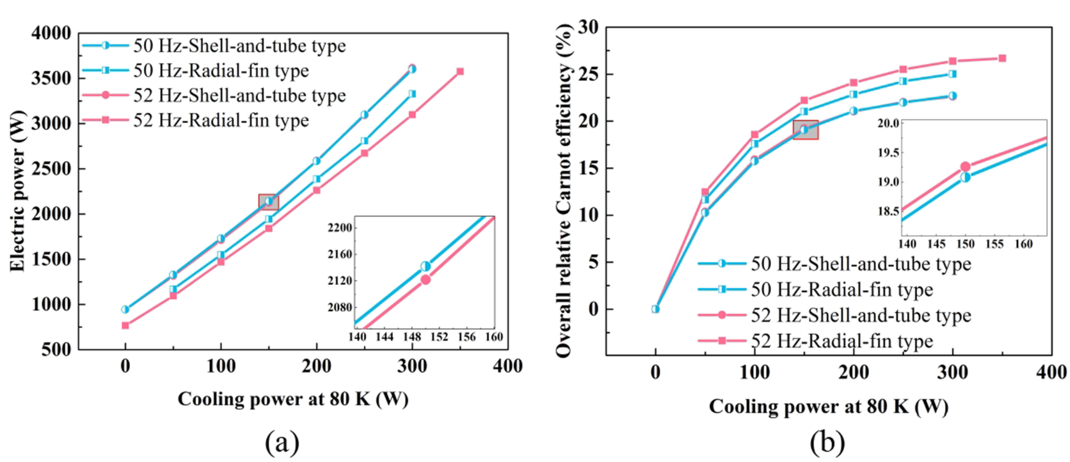


Fig. 16. Dependence of the (a) consumed electric power and (b) overall relative Carnot efficiency on cooling power at 80 K for the two type of AHXs.

the heat transfer performance of the shell-and-tube type AHX, thus a better cooling performance of the FPSC could be well anticipated.

4. Conclusions

In this work, operating characteristics of a refurbished high cooling capacity free-piston Stirling cryocooler were studied both numerically and experimentally. The effect of key operating parameters including mean pressure and working frequency as well as AHX configuration were numerically investigated to find the optimal values and configuration. Based on the simulations, a series of experiments were carried out to verify the optimization.

For the current design, there is an optimal mean pressure and operating frequency to achieve the best cooling performance of the cooler. With the aid of numerical optimization, a validation has been conducted on the improved experimental setup. The experimental results show that when the mean pressure is 2.75 MPa and working frequency is 52 Hz, the free-piston Stirling cryocooler is capable of providing 350 W cooling power at 80 K with an electric power input of 3.57 kW, and the relative Carnot efficiency of the overall system reaches up to 26.8%. Compared with the previous work, the cooling performance is greatly improved and the deviations between the experimental and the numerical results on cooling performance are successfully reduced. A further attempt of improvement has been done on the ambient heat exchanger. A preliminary test on an off-the-shelf shell-and-tube type ambient heat exchanger exhibits a lower performance compared with the current radial-fin type ambient heat exchanger, however, showing a good agreement with the calculations. A better performance could be well expected with an optimized shell-and-tube type ambient heat exchanger.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is financially supported by the National Key Research and Development program of China (No. 2016YFB0901403 and 2016YFE0102200), National Natural Science Foundation of China (No. 51876214).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.applthermaleng.2020.115101.

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